SOCIAL SCIENCE -II

STANDARD - IX

CHAPTER – 3

PLATEAU WHERE THE EARTH'S HISTORY SLUMBERS

@ Which major physiographic division of India is introduced in this chapter?

- The physiographic division introduced is the Peninsular Plateau

@ What are the shape and average altitude of the Peninsular Plateau?

- It is an almost triangular-shaped physiographic unit.
- Average altitude of 600 to 900 metres above mean sea level.

@ Why is this physiographic unit called the "Peninsular Plateau"?

- It is named so because this division holds the major share of Peninsular India.

@ What are the key physical diversities of the Peninsular Plateau

- Extensive table lands with bordering mountain ranges and hills
- Comparatively shallow river valleys

@ What is a Plateau?

- Plateaus are the relatively flat and very extensive landforms situated at comparatively higher elevation from the surroundings.

@ What are the three types of plateaus classified based on their location?

There are three types of plateaus based on their location:

- Intermontane plateaus
- Piedmont plateaus
- Continental plateaus

@ What are the boundaries, total area, and classification of the Peninsular Plateau?

- The Peninsular Plateau lies south of the North Indian Plain bordered by the Western Ghats in the west and the Eastern Ghats in the east.
- This landform extending over 16 lakh sq.km
- Based on location, it is generally classified into two:

i. The Deccan Plateau

ii. The Central Highlands

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The Deccan Plateau

@ Describe the location and northern boundaries of the Deccan Plateau.

- It lies south of the Satpura ranges between the Western Ghats and the Eastern Ghats.
- Satpura ranges, Maikala ranges and the Mahadeo Hills form the northern boundary of Deccan Plateau.

@ What is the linguistic origin of the word "Deccan"?

- The term 'Deccan' has been derived from the Sanskrit word 'Dakshin', meaning 'the South'.

@ Which rock types make up the Deccan Plateau, and how were they formed?

- The Deccan Plateau is made up of crystalline rocks, like **basalt, granite and gneiss,** formed by the lava flows millions of years ago.

@ What is the "Deccan Trap"?

- The north western part of the Deccan Plateau is composed of lava rocks called basalt. This region is called as Deccan Trap.

@ What is "Regur soil"? Detail its properties and agricultural significance.

- The black soil formed by the weathering of basalt rocks is the peculiarity of this region.
- This soil is also known as Regur soil.
- Being highly fertile and with more water - retaining capacity, it protects the agricultural crops even in summer.
- This soil is also called as black cotton soil, as it is very useful for cotton cultivation.
- Minerals like lime, iron, magnesium and aluminium are characteristics of regur soil.

@ What are the main features of the Western Ghats?

- Extending for about 1600 km from Kanyakumari in the south to the state of Gujarat in the north
- This range is the western edge of the Deccan Plateau.
- The height of this range gradually increases from north to south.
- Anamudi , the highest peak in Peninsular India, is located in Anamalai of the Western Ghats .
- Most of the peninsular rivers originate from the Western Ghat ranges.

@ What are the different regional names of the Western Ghats?

- The Western Ghat ranges, known as Anamalai and Elamalai in Kerala, is called as the Nilgiris in Karnataka and Tamilnadu, and the Sahyadris, in Maharashtra.

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@ What are the main features of the Eastern Ghats?

- Comparatively lower in height than the Western Ghats
- The total extent of the Eastern Ghats is about 800 km from the Mahanadi banks in Odisha to the Nilgiri ranges in Tamil Nadu.
- The east-flowing peninsular rivers cut across the Eastern Ghats, breaking the continuity of these ranges and flow over the eastern coastal plains.
- The Western Ghats and the Eastern Ghats join at the Nilgiri Hills.

Central Highlands

- Central Highlands are the extensive plateau region that lies to the north of Satpura ranges.
- The Aravali mountains are on the western margin of this table land known as **Malwa Plateau.**

@ What are the characteristics of the Aravali ranges?

- The Aravali ranges is an example for old fold mountains or residual mountains worn down by long term erosion.
- Mount Abu, a major hill station, is in Aravali ranges.
- Mount Abu is also the highest peak in Malwa Plateau.

@ Which plateau region lies along the eastern part of the Central Highlands, and what is its significance?

- The plateau region along the eastern part of the Central Highlands is the Chota Nagpur Plateau.
- Richest store house of minerals.
- Metallic minerals such as iron ore, bauxite, manganese and copper and non - metallic minerals such as lime stone and coal make this region mineral rich.

Climatic Diversity in Peninsular Plateau

@ What type of climate does the Peninsular Plateau generally experience?

- Peninsular Plateau generally experiences tropical monsoon climate, though there is considerable spatio-temporal variation in temperature and rainfall.

@ What are the five major factors that influence the climate of the Peninsular Plateau?

- Tropical location
- Peculiar shape of the Peninsula
- Distance from the ocean
- Orientation of mountain ranges
- Direction of Monsoon winds

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@ Describe the summer temperature patterns across the Peninsular Plateau.

- Summer temperature in the Peninsular Plateau is more than 30 Degree Celsius.
- Temperature at the Deccan Plateau generally touches 38 Degree Celsius in March.
- Generally low temperature prevails at higher elevations in the Western Ghats.

@ What is the "diurnal range of temperature," and why is it very high in the plateau's interiors?

- Diurnal range of temperature is the difference between the maximum temperature and minimum temperature recorded at a particular place in a day.
- The diurnal range of temperature is very high along the interiors of the plateau due to the **considerable fall in the night temperature**.

@ Why does the Peninsular Plateau receive less rainfall than the western slopes of the Western Ghats?

- Rainfall is moderate or scanty throughout the Peninsular Plateau except the western slopes of the Western Ghats.
- During the southwest monsoon season the moisture – laden winds, raised and condensed along the western slopes of the Western Ghats cause heavy rainfall along the windward side.
- The winds descending along the eastern slopes of the Western Ghats being dry, the plateau regions situated close to the eastern slopes receive very less amount of rainfall
- Such regions are termed as rain **shadow regions.**

@ Describe the weather conditions on the plateau during the northeast (retreating) monsoon season.

- Peninsular Plateau generally experiences dry climate during the northeast monsoon or retreating monsoon season.
- Though the low pressure whirls developed over the Bay of Bengal causes heavy showers along the east coast, especially Tamil Nadu and Andhra Pradesh, the plateau remains unaffected.

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Peninsular Rivers

@ **Catchment Area:** The defined area from where the water flows into a river is termed as the catchment area of the river.

@ **Drainage Basin**: The area formed by a river and its tributaries is called a Drainage Basin.

@ **Water Divide**: The boundary line separating two watersheds or drainage basins is called Water Divide.

@ How is the drainage system of Peninsular India divided based on the major water divides?

- The Western Ghats, the mountain ranges of the Central Highlands and the Aravali ranges extending up to the Delhi ridges, divides the peninsular drainage into three:
  I) The peninsular rivers flowing eastwards into the Bay of Bengal

II) The Peninsular rivers flowing westwards into the Arabian Sea.
III) Rivers that flow northwards to join Yamuna and Ganga.

@ What are the common characteristics of the east-flowing peninsular rivers?

- Originate from the Western Ghats.
- They cut across the Peninsular Plateau and flows eastwards through the eastern coastal plain to join the Bay of Bengal.
- Rivers are more seasonal in their flow.

@ Major East-flowing Peninsular Rivers

River	Source	Major Tributaries	States sharing the River Basin
Mahanadi	Sihawa in Raipur (Chattisgarh)	Ib, Tel	Chattisgarh, Odisha, Madhya Pradesh
Godavari	Nashik in Maharashtra	Pranhita, Indravati, Sabari	Maharashtra, Madhya Pradesh, Andhra Pradesh
Krishna	Mahabaleshwar in Maharashtra	Tungabhadra, Bhima, Koyna	Maharashtra, Karnataka, Andhra Pradesh
Kaveri	Brahmagiri Hills in Karnataka	Kabani, Bhavani, Amaravati	Karnataka, Tamilnadu, Kerala

@ Why does the water flow in River Kaveri not experience any significant decrease throughout the year compared to other peninsular rivers?

- This is due to the fact that the catchment area of the River Kaveri receives Southwest Monsoon rainfall during summer and Northeast Monsoon rainfall during winter.

West-flowing Peninsular Rivers

- Most of the peninsular rivers, except Narmada and Tapti, originate from the western slopes of the Western Ghats and swiftly flow into the Arabian Sea through the western coastal plains.

River	Source	Major Tributaries	States holding the river basin
Narmada	Amarkantak (Madhya Pradesh)	Hiran, Banjar	Madhya Pradesh, Maharashtra, Gujarat
Tapti	Multai (Madhya Pradesh)	Purna, Girna	Madhya Pradesh, Maharashtra, Gujarat

Peninsular Rivers joining River Ganga

- Rivers originating from the Malwa Plateau flowing northward either to join River Yamuna or directly to River Ganga.
- The major Peninsular rivers that join the River Ganga are: Yamuna, Chambal, Betwa, Ken, Son

@ Peninsular rivers are, in general, not navigable. Why?

- Peninsular rivers are more seasonal in their flow. The water flow decreases during summer,making them less suitable for navigation.

@ What are Multipurpose River Valley Projects?

- Multipurpose River Valley projects are projects that construct dams across a river for serving different purposes simultaneously.
- Flood control, irrigation, hydel-power generation, inland water transport, fishing and tourism are some of the main objectives of such projects.

@ Multi-purpose river valley projects in the Peninsular India.

River Valley Project	River	State
Hirakud	Mahanadi	Odisha
Thungabhadra	Thungabhadra (Tributary of Krishna)	Karnataka
Sardar Sarovar	Narmada	Gujarat
Krishna Raja Sagar	Kaveri	Karnataka
Nizam Sagar	Godavari	Andhra Pradesh

Natural Vegetation in the Peninsular Plateau

@ What are the major natural vegetation types in the Peninsular plateau?

I) Tropical Deciduous Forests

- These are the most widespread natural forests in the Peninsular Plateau.
- Based on the availability of rainfall, these are of two types:
  i) Moist deciduous forests
  ii) Dry deciduous forests

Moist deciduous forests

- Found in areas receiving 100–200 cm annual rainfall.
- Common along the eastern slopes of the Western Ghats.
- Also found in Madhya Pradesh, Chhattisgarh and Chota Nagpur.
- Common trees: Teak, Sal, Shisham, Mouva and Sandalwood.

Dry deciduous forests

- Found in areas receiving 70–100 cm annual rainfall.
- Common in other parts of the Peninsular Plateau.
- Changes into thorn forests and shrubs as rainfall decreases.
- During drought, trees shed their leaves completely.
- Forests may turn into grasslands with leafless trees.
- Common trees: Teak, Rosewood, Axle wood and Bamboo.

Tropical Thorn Forests

- Found in high-temperature regions.
- Annual rainfall: below 75 cm.
- Short trees and thorny shrubs are common.
- Main vegetation: Acacia, Euphorbia, Date palms and grasses.
- Found in semi-arid Maharashtra and Karnataka (east of Western Ghats).
- Also found in dry regions of Andhra Pradesh, Telangana and Tamil Nadu.

Southern Montane Forests

- Found in the higher reaches of the Peninsular Plateau.
- Major areas: Western Ghats, Vindhya Ranges, Nilgiri Hills.
- Above 1500 m: Temperate vegetation.
- Below 1500 m: Sub-tropical vegetation.
- Sub-tropical vegetation in Nilgiri, Palani and Anamalai hills is called Shola Forests.

Soil Types in Peninsular Plateau

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- Most of the soil types found in the Peninsular Plateau are in- situ soils.
- The soils found here can be classified as black soil, red soil, laterite soil and mountain forest soils.

@ Differentiate between "In-situ soils" and "Transported soils" with examples.

- In-situ soils are the soils that rest over the place at which they are formed.
  Example: Black soil
- The soils carried away by rivers, wind etc. and deposited somewhere else are called Transported soils. Example: Alluvial soil

Black Soil

@ How is Black Soil formed, what are its other names, and where is it distributed?

- Formed by the weathering of basalt (lava rocks).
- Mainly found in Maharashtra and Madhya Pradesh.
- Also found in parts of Karnataka, Telangana, Andhra Pradesh, Gujarat and Tamil Nadu.
- Other names: Regur Soil / Black Cotton Soil.

Red Soil

@ Explain the formation and color characteristics of Red Soil.

- Formed by the weathering of old crystalline metamorphic rocks.
- Mainly found in the Peninsular Plateau.
- Colours: Red, brown, grey and yellow.
- Red colour is due to the presence of iron.

Laterite Soil

@ Describe the formation, distribution, and agricultural utility of Laterite Soil.

- Formed due to leaching of minerals like silica and lime.
- Found mainly along the Western & Eastern Ghats, Rajmahal Hills, Vindhya, Satpura and Malwa Plateau.
- Less fertile and generally not arable.
- With fertilisation, suitable for tea, coffee, rubber and arecanut plantations.

Mountain Soil

@ Where is mountain soil found in South India, and which crops is it suitable for?

- Found along the Western and Eastern Ghats of South India.
- Suitable for plantation crops.
- Main crops: Tea, coffee, spices and tropical fruits.
- Mainly found in Karnataka, Tamil Nadu and Kerala.

Agriculture in Peninsular Plateau

@ Why is agriculture possible only in selected regions of the Peninsular Plateau?

- Undulating topography
- Eroded surface soil
- Steep slopes
- Thin topsoil
- Exposed rocks
- Scattered hills

@ What are the major crops cultivated in the Peninsular Plateau?

- Rice, wheat, cotton, sugarcane and tobacco
- Plantation crops: Tea and coffee

@ What are the major features of agriculture in the Western Ghats?

- Plantation crops dominate.
- Tea and coffee are common in the Nilgiris.
- Paddy is cultivated using hill terraces

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@ Which Indian states lead in coffee production, and which varieties of coffee are primarily grown?

- Karnataka is the leading coffee producer in India.
- Karnataka has about 59% of India's coffee plantations.
- It contributes about 71% of India's coffee production.
- Kerala ranks second with about 22% production.
- Major varieties: Arabica and Robusta.

@ Explain the distribution and importance of tea cultivation in the Peninsular Plateau.

- Tea plantations are mainly found in Nilgiri Hills and Western Ghats.
- Spread over Tamil Nadu, Karnataka and Kerala.
- The region has about 44% of India's tea plantations.
- It contributes about 25% of India's total tea production.
- Tea cultivation is labour-intensive.
- Provides employment in plantations and allied industries.

@ What are the favourable conditions for sugarcane cultivation in the Deccan Plateau?

- Black lava soil in the Deccan Plateau
- Tropical climate and long crushing season
- Comparatively high sucrose content in the tropical variety of sugarcane

@ Explain the planting and harvesting seasons of cotton in Peninsular India, along with its climatic and soil requirements, production and distribution.

- Planting: Begins in October.
- Harvesting: From January to May.
- 7 frost-free months are essential during the early growth period.
- Temperature: 21°C–30°C.
- Annual rainfall: 50–100 cm.
- Can be grown in low-rainfall areas with irrigation.
- Black soil of the Deccan–Malwa Plateau is most suitable.
- Gujarat is the leading producer, followed by Maharashtra.

Storehouse of Minerals

@ Why is the Peninsular Plateau called a storehouse of minerals?

- Mineral resources are largely concentrated in the Peninsular Plateau.
- Chota Nagpur Plateau is called the heartland of minerals.
- Chotanagpur–Odisha Plateau is the richest mineral belt in India.
- Important minerals: Coal, iron ore, manganese, mica, bauxite and copper.

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@ Explain the major mineral zones of the Peninsular Plateau and mention the important minerals found in each zone.

Major Mineral Zones of the Peninsular Plateau

Mineral Zone	Major Minerals
Northeastern Region	Coal, Iron ore, Manganese, Mica, Bauxite, Copper
Central Region	Manganese, Bauxite, Limestone, Marble, Coal, Mica, Iron ore, Graphite
Southern Region	Iron ore, Bauxite, Lignite
Southwestern Region	Iron ore, Clay
Northwestern Region	Copper, Lead, Zinc, Uranium, Mica

Life in Peninsular Plateau

@ Explain the factors that influenced the growth of population and settlements in the Peninsular Plateau.

- The Peninsular Plateau is moderately populated.
- In the earlier stages, settlements were limited due to non-arable topography and continental climate.
- beginning of mining activities, development of road-rail network and emergence of mineral based industries, people were attracted to the plateau.
- Scope of commercial agriculture based on irrigation and technological possibilities also led to an increase in population here.

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